# Problemset 3 AMAS

Asger Thormann

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# 1

## 1.1

I use pandas to load the .csv file. I find that in the column 'earning' there are 1161 entries of ' $> 50K$' and 3639 entries of '$<= 50K$'. I change ' $> 50K$' to 1 and '$<= 50K$' to 0 to use for training. I ensure that there 500 entries of high earning left for the test and I try to keep the structure of the data set by ensuring that the ratio of high earners to low earners are the same for the training data as in the full data set. I sample the training subset randomly and don't reuse these for testing.

Table 1 shows the train/test split:

| **Dataset** | **High** | **Low** | **Total** |
|---|---|---|---|
| Train | 661 | 2071 | 2732 |
| Test | 500 | 1568 | 2068 |

**Table 1:** Data points used for training and testing

To optimize the hyper parameters I do a randomized grid search using 400 random combinations.

I find the accuracy using the test data set to be:

$$a = 80.03\%$$

## 1.2

Using 500 of each category I plot the distribution of the test statistic, which here is probability of being a "high earner". The distribution is shown in figure 1.

To make the best decision on when to classify a data point as "high" or "low" I calculate and implement a best threshold. From the ROC curve I find the True Positive Rate and False Positive Rate and calculate:

$$J = TPR - FPR$$

I use the index where this is maximized as threshold. This gives me a new accuracy:

$$a = 71.08\%$$

Using this threshold I classify the blind data. This is shown in figure 2

I include the files with identified "high earners" and "low earners"

## 1.3

The training data consists of the following:
The fraction of high earners is $f_{high} = 0.24$.
The fraction of low earners is $f_{low} = 0.76$.

## 1.4

I try to avoid overfitting by evaluating the accuracy of on the training dataset and comparing this to the accuracy on the test dataset. I find

$$a_{train} = 0.7339$$

$$a_{test} = 0.7108$$

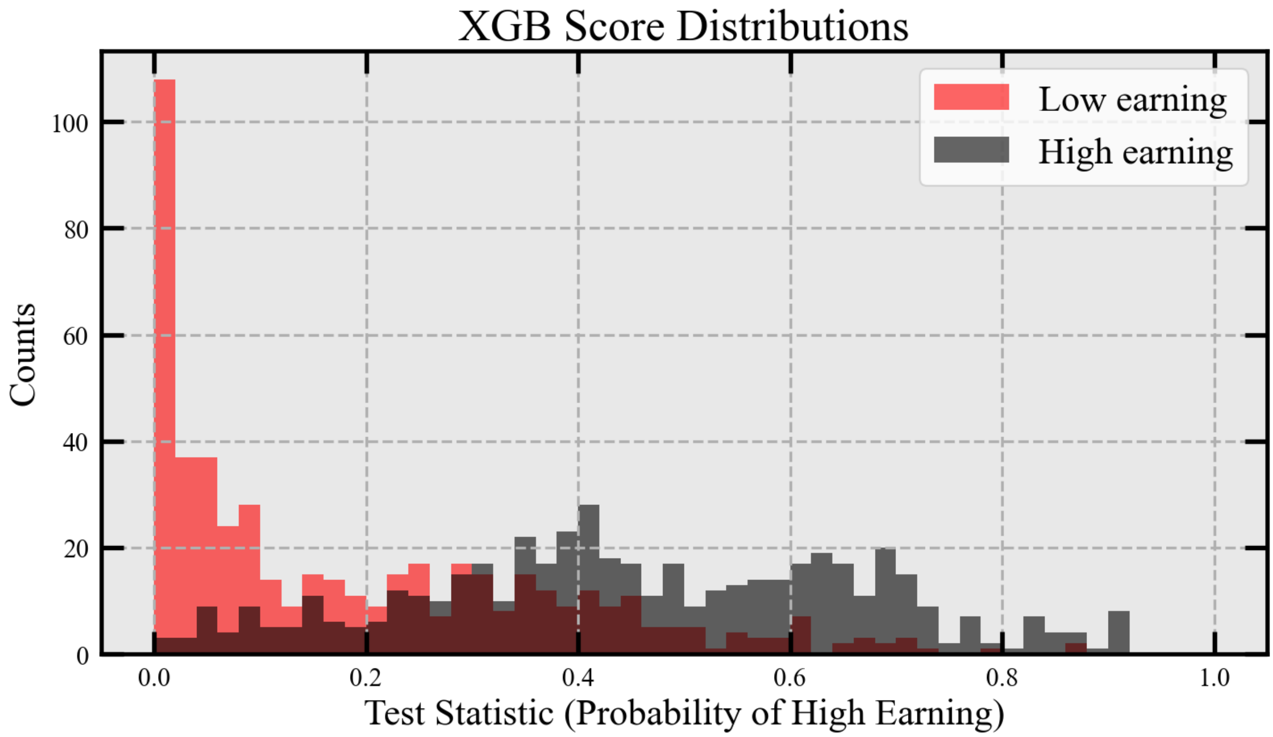


**Figure 1:** Distributions of scores. Red is known "low earners" and black is known "high earners"

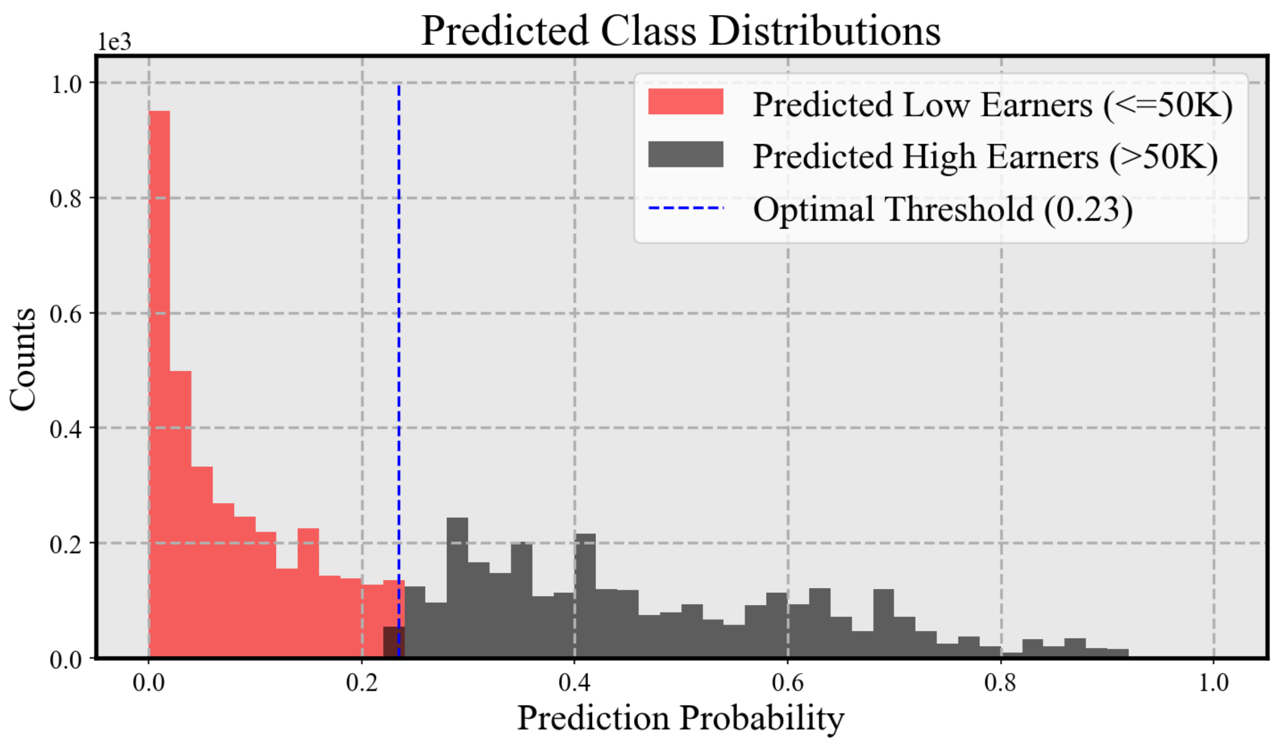


**Figure 2:** Classification of blind data using the optimized threshold from the test data.

As these are fairly close i can conclude that the model hasn't memorized the training data as these values would deviate more. I avoid using many more instances of "high earners" in my training subset as this could result in the model learning that these are more frequent. I train the model using the same frequency of high and low as in the total dataset, thus assuming that the blind data also follows this (which might not be true). I can also see from my hyperparameters that shallow trees are chosen as well as a slow learning rate. This is a good indication of not overfitting.

## 2

### 2.1

I plot each crash in the dataset as a function of the latitude and longitude. This is shown in figure 3. I use the keyword 'lat' and 'lon' as specified in the problem set.

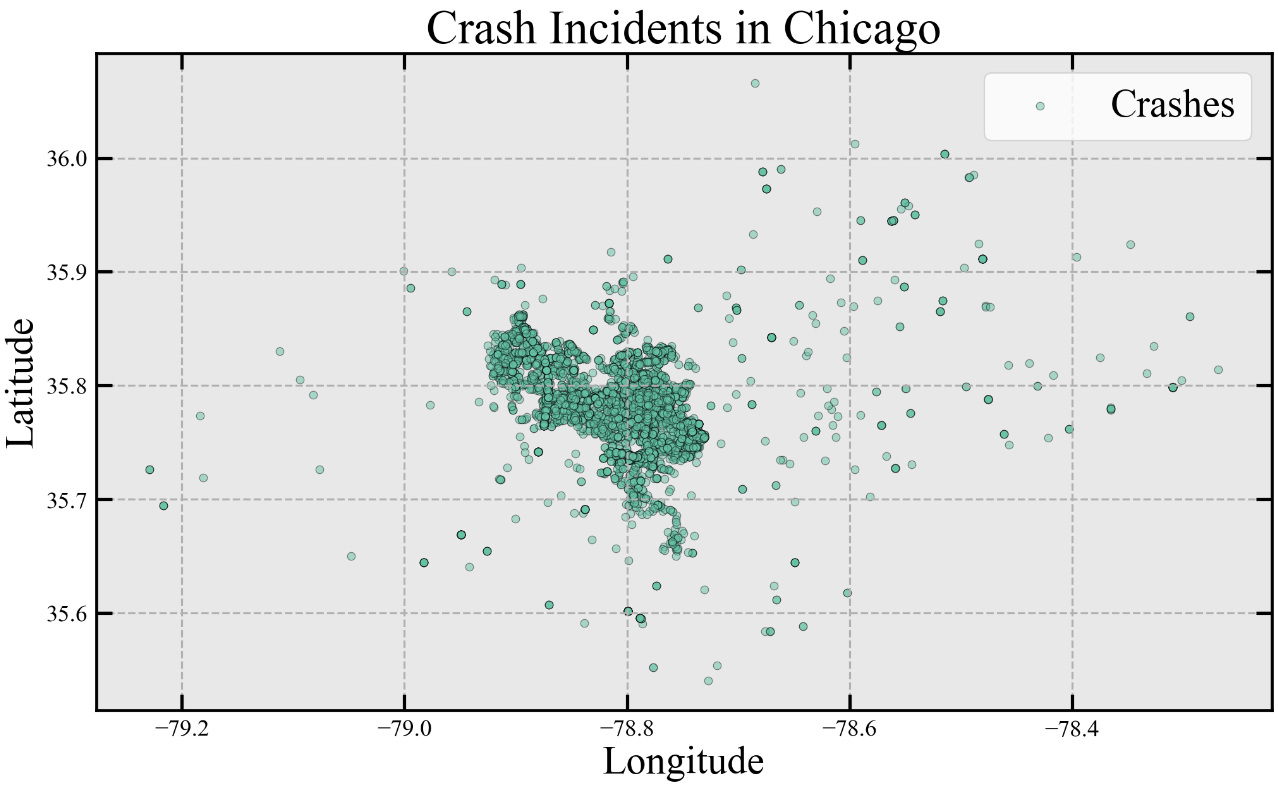


**Figure 3:** Each crash in the dataset.

### 2.2

When plotting the times I use the label 'crash_date' as this is given in universal standard time I convert it to Chicago time. The histogram with a bin width of 1 hour is shown in figure 4

### 2.3

To create a Gaussian KDE with a bandwidth of 0.25 it is important to address the periodicity of the problem. As boundary times are close in reality the probability of points near the edges contribute on the other side of midnight. To account for this one can copy the dataset and thus have replicated data before and after.
The number of replicated data sets at both ends can be chosen based on the band width of the problem.

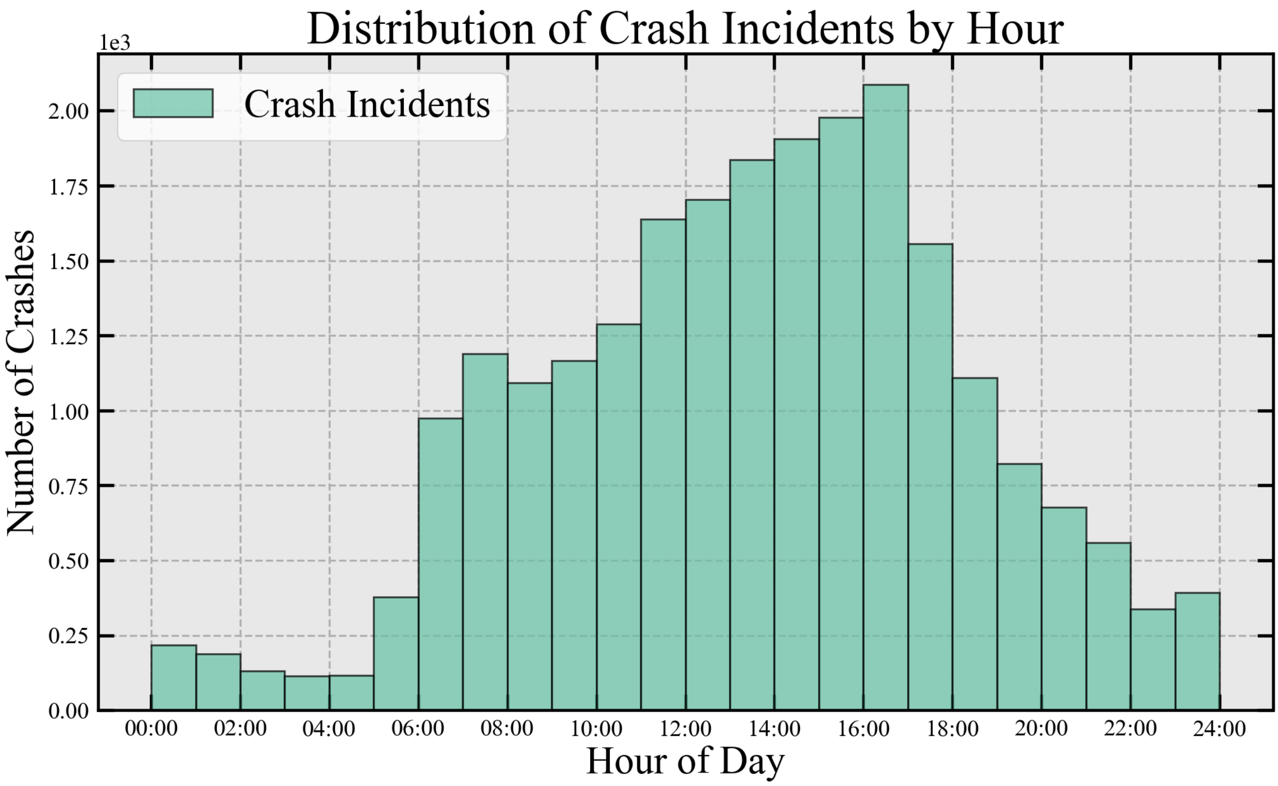


**Figure 4:** Crash times adjusted for time zone of Chicago

It is important to normalize correctly as the KDE should still integrate to 1 over the 24 hour period. In figure 5 I illustrate how data replication can be implemented. I've constructed gaussians around 0 and 24. The blue line shows replication of the datasets. This is scaled to integrate to 1 in the range $t = [0, 24.00]$.

### 2.4

From sklearn.neighbors I use the Epanechnikov kernel on the dataset. I set the bandwidth to 0.8. I use the full dataset by converting minutes and seconds to hours in decimals. Figure 6 show the PDF.
I use the periodic method described above by duplicating the dataset. Using this method I need to scale the pdf with a factor of 3 as i use 3 datasets. This means that the KDE PDF integrates to 1 over the range $x = [0, 24.0]$.

### 2.5

I convert the times to decimal values as shown in table 2. I know I have found different values than my fellow students but I have been unable to find the error and as the pdf does indeed integrate to one I don't know what to check.

### 2.6

To choose the best 2-hour time interval for additional policemen to patrol, I scan across the KDE with a 2-hour window, integrating the region. I handle the periodic boundary while doing this. I save the 2-hour window containing the largest cumulated probability. Usimg this method i find that the best time window is 14:58:01 - 16:58:01. Looking at the dataset and the KDE PDF this looks true. The reduction in this period is 1.71%

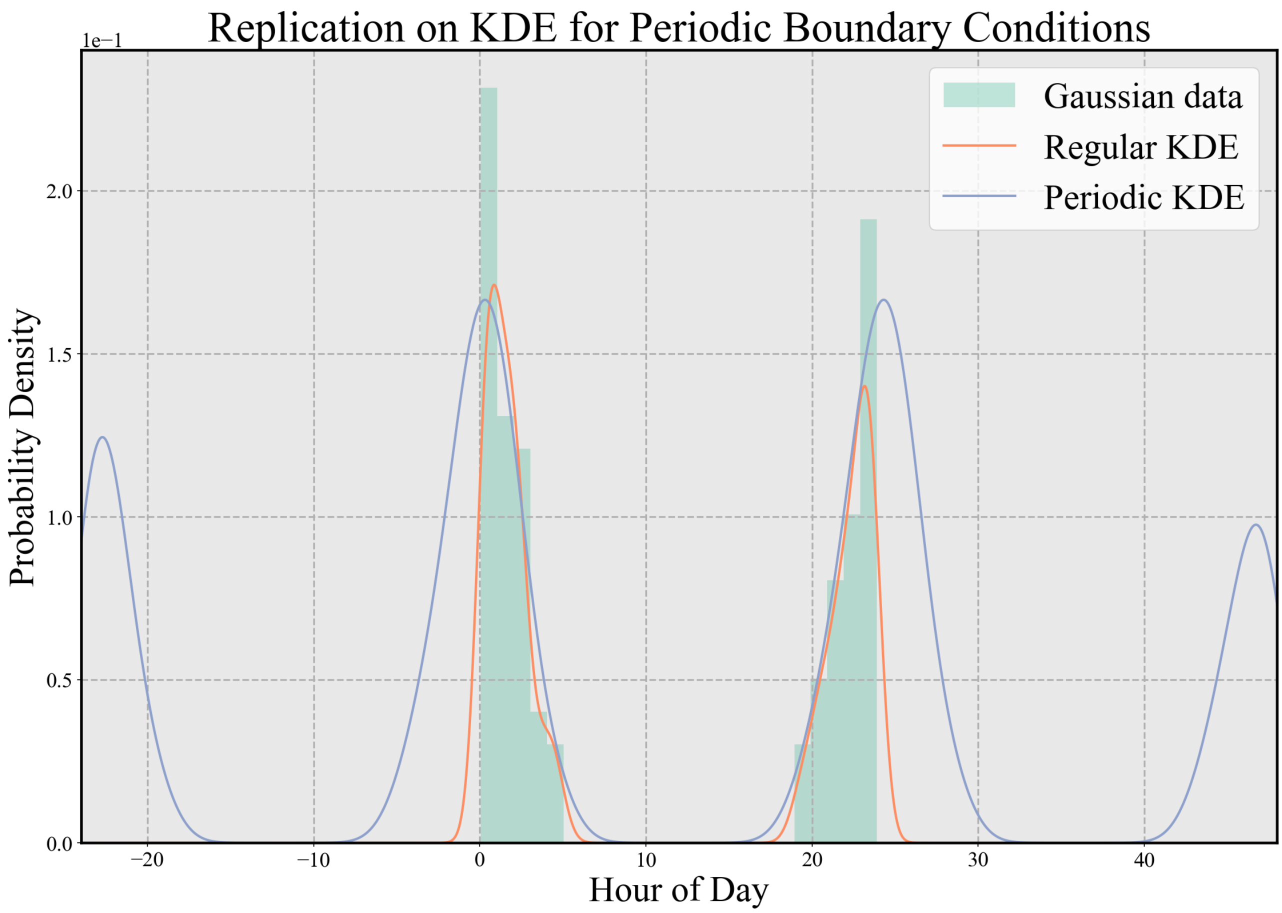


**Figure 5:** Gaussian data sampled with $\mu_1 = 0, \sigma_1 = 2$ and $\mu_2 = 24, \sigma_2 = 2$

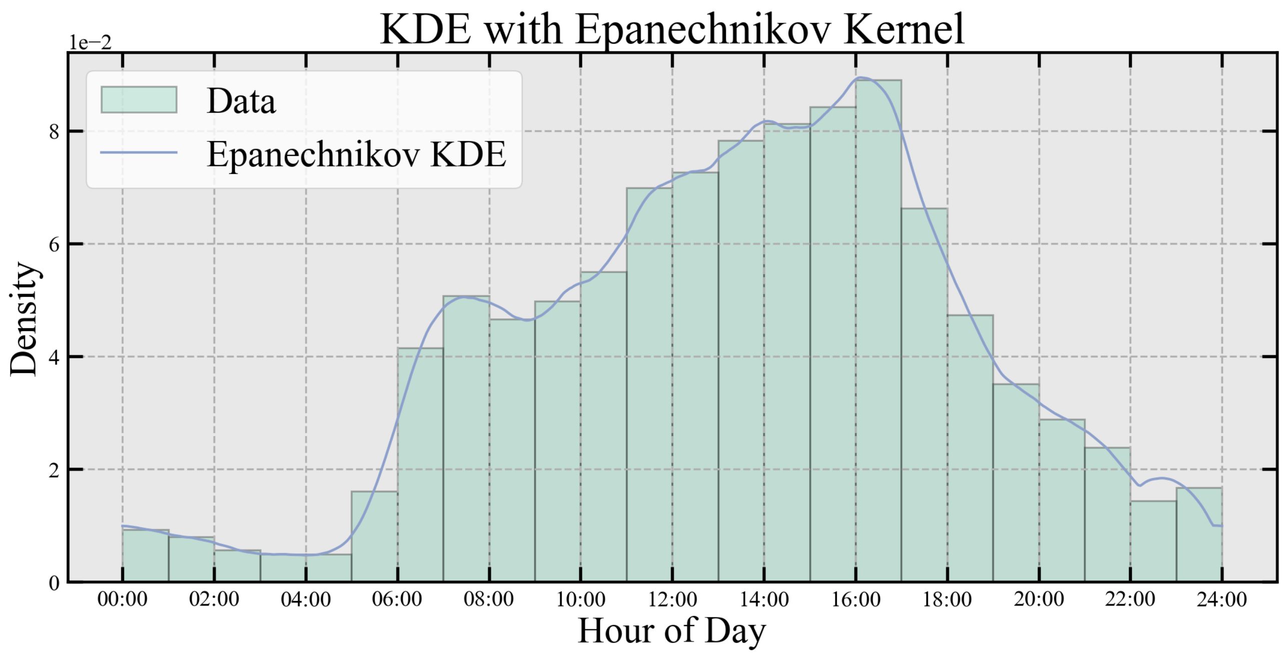


**Figure 6:** Epanechnikov KDE

| Time (hours) | Density |
|---|---|
| 0.3833 | 0.00949 |
| 1.8167 | 0.00732 |
| 8.2000 | 0.04880 |
| 15.9167 | 0.08842 |
| 18.0333 | 0.05587 |
| 21.2000 | 0.02569 |
| 23.7333 | 0.01097 |

**Table 2:** Epanechnikov KDE at given times

### 2.7

I find that many of the longitude and latitude values are NaN. I filter these out. I see that there is also columns with labels "lat2" and "lon2" I don't use these, mainly because they seem very off but also because the problem set says "use 'lat' and 'lon'".

Again I use the package like earlier and I plot on af (500x500) grid for the contour map. This is shown in figure 7. I've checked that the KDE integrates to $\approx 1$ (0.9996) over the entire domain.

### 2.8

I integrate the 2D KDE inside the white box shown in figure 7. This evaluates to: $f_{box} = 0.137$

## 3

### 3.1

To find the probability of a defect pacemaker coming from A2 I use Bayes theorem by inserting values from the table:

$$P(A2; D) = \frac{P(D; A2) \cdot P(A2)}{P(D)} = 18.3\%$$

Here I calculate the marginal likelihood as $P(D) = \sum_i P(D; A_i) \cdot P(A_i)$

### 3.2

I repeat the calculation for all factories and obtain the values shown in table 3. It is shown that A5 is most likely.

| Cell | Posterior probability (%) |
|---|---|
| A1 | 21.4 |
| A2 | 18.3 |
| A3 | 15.3 |
| A4 | 21.4 |
| A5 | 23.7 |

**Table 3:** Probabilities for factories

### 3.3

To make it equally likely for a defect pacemaker to originate from any factory the posterior becomes $P(A_i; D) = \frac{1}{5}$. Further more the defective rates can only increase for each factory. I write the the

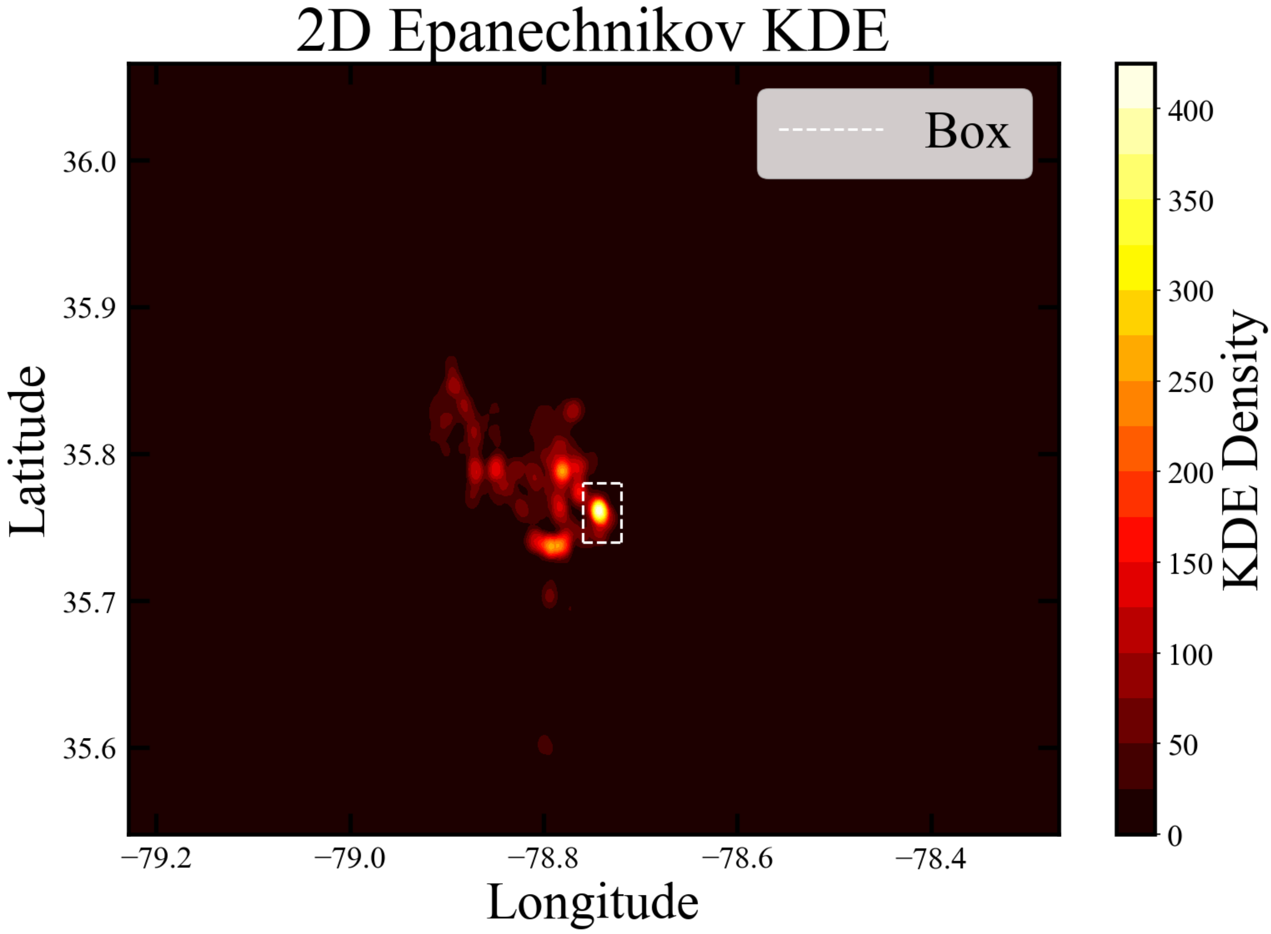


**Figure 7:** Contour density of the KDE over all car crashes

posterior as

$$\frac{P(D; A_i) \cdot P(A_i)}{\sum_i P(D; A_i) \cdot P(A_i)} = \frac{1}{5}$$

Which implies that $P(D; A_i) \cdot P(A_i) = c$ where $c$ is an constant. Using this in the marginal likelihood:

$$\sum_i P(D; A_i) \cdot P(A_i) = \sum_i c = i \cdot c = 5c$$

as we have 5 factories. Rewriting this i have:

$$c = \frac{P(D)}{5}$$

Now the constraint that defective rate only can increase can be implemented to make new defective rates:

$$\frac{c}{P(A_i)} = P(D; A_i)_{new}$$

where:

$$P(D; A_i)_{old} \leq P(D; A_i)_{new}$$

This results in the minimum value of c being:

$$c_{min} = max(P(A_i) \cdot P(D; A_i)_{old}) = 0.00775$$

The values are shown in table 4.

| Cell | Defective (%) |
|---|---|
| A1 | 2.2 |
| A2 | 5.2 |
| A3 | 15.5 |
| A4 | 3.9 |
| A5 | 3.1 |

**Table 4:** Updated defective rates

I check the posterior using these production rates and they are equal to 0.2 for every factory.

## 4

Unfortunately I haven't had the time to do part 4 of this assignment.